

Trabalho

2025-07-09

Análise dos dados

```
load("bfd_2022.rdata")
```

Abaixo o summary da base de dados:

```
summary(bfd)
```

```
##      arrival            depart            route            company
## Length:845160      Length:845160      Length:845160      Length:845160
## Class :character    Class :character    Class :character    Class :character
## Mode  :character    Mode  :character    Mode  :character    Mode  :character
##
##
##
##      flight            di            type
## Length:845160      Length:845160      Length:845160
## Class :character    Class :character    Class :character
## Mode  :character    Mode  :character    Mode  :character
##
##
##
##      depart_day_period      arrival_day_period      expected_depart
## Night      : 85081      Night      :100822      Min.      :2022-01-01 00:05:00
## Early Morning:165463      Early Morning:137982      1st Qu.:2022-04-09 07:45:00
## Mid Morning  : 94069      Mid Morning  : 84877      Median :2022-07-13 05:00:00
## Late Morning : 87431      Late Morning : 97092      Mean   :2022-07-08 23:42:38
## Afternoon    :165178      Afternoon    :178550      3rd Qu.:2022-10-08 18:31:15
## Early Evening:141867      Early Evening:129243      Max.   :2023-01-01 21:30:00
## Late Evening :106071      Late Evening :116594
##      real_depart            expected_arrival
## Min.      :2021-12-31 23:50:00      Min.      :2022-01-01 01:20:00
## 1st Qu.:2022-04-08 19:03:00      1st Qu.:2022-04-09 09:50:00
## Median :2022-07-12 13:45:00      Median :2022-07-13 07:25:00
## Mean   :2022-07-08 10:20:13      Mean   :2022-07-09 02:08:51
## 3rd Qu.:2022-10-07 14:58:00      3rd Qu.:2022-10-08 21:06:15
## Max.   :2023-01-01 20:07:00      Max.   :2023-01-02 09:25:00
## NA's    :34984
##      real_arrival            status_depart            status_arrival
## Min.      :2022-01-01 01:05:00      Length:845160      Length:845160
## 1st Qu.:2022-04-08 21:06:00      Class :character    Class :character
## Median :2022-07-12 15:59:00      Mode  :character    Mode  :character
## Mean   :2022-07-08 12:42:06
## 3rd Qu.:2022-10-07 17:05:00
```

```

## Max.      :2023-01-02 00:26:00
## NA's      :34984
## observation      delay_depart      delay_arrival      expected_flight_length
## Mode:logical    Min.      :-2040.000    Min.      :-2032.00    Min.      : 10.0
## NA's:845160     1st Qu.:   -7.000    1st Qu.:  -11.00    1st Qu.:  75.0
##               Median :   -2.000    Median :   -4.00    Median : 100.0
##               Mean   :    5.356    Mean   :    2.57    Mean   : 146.2
##               3rd Qu.:    6.000    3rd Qu.:    7.00    3rd Qu.: 165.0
##               Max.   :44873.000    Max.   :44838.00    Max.   :1400.0
##               NA's   :34984      NA's   :34984
## real_flight_length outlier_depart_delay outlier_arrival_delay
## Min.      : 0.0      Mode :logical      Mode :logical
## 1st Qu.: 73.0      FALSE:809858      FALSE:809867
## Median : 99.0      TRUE :318          TRUE :309
## Mean   : 141.9      NA's :34984        NA's :34984
## 3rd Qu.: 159.0
## Max.   :2285.0
## NA's   :34984
## outlier_expected_flight_consistency outlier_real_flight_consistency
## Mode :logical      Mode :logical
## FALSE:845160        FALSE:810176
##                     NA's :34984
##
##
##
## outlier_expected_flight_length outlier_real_flight_length
## Mode :logical      Mode :logical
## FALSE:815630        FALSE:805264
## TRUE :29530         TRUE :4912
##                     NA's :34984
##
##
##
## depart_air_temperature depart_dew_point depart_relative_humidity
## Min.      : 1.0      Min.      :-7.0      Min.      : 9.72
## 1st Qu.:19.0      1st Qu.:14.0      1st Qu.: 62.03
## Median :23.0      Median :18.0      Median : 74.44
## Mean   :22.9      Mean   :17.3      Mean   : 73.29
## 3rd Qu.:27.0      3rd Qu.:21.0      3rd Qu.: 88.09
## Max.   :45.0      Max.   :33.0      Max.   :100.00
## NA's   :83934      NA's   :84997      NA's   :84997
## depart_wind_direction depart_wind_speed depart_sky_coverage depart_pressure
## Min.      : 0.0      Min.      : 0.00    Length:845160    Min.      : 0.50
## 1st Qu.: 80.0      1st Qu.: 4.00    Class :character  1st Qu.:29.91
## Median :130.0      Median : 6.00    Mode  :character  Median :30.00
## Mean   :151.1      Mean   : 6.79                    Mean   :30.00
## 3rd Qu.:200.0      3rd Qu.: 9.00                    3rd Qu.:30.09
## Max.   :360.0      Max.   :82.00                    Max.   :32.72
## NA's   :112521     NA's   :83674                    NA's   :83718
## depart_visibility depart_apparent_temperature      depart_wind_speed_scale
## Min.      :0.000    Min.      :-1.183      Light Breeze      :244460
## 1st Qu.:6.210      1st Qu.:19.000      Gentle Breeze     :239504
## Median :6.210      Median :23.000      Light Air         :121902

```

```
## Mean :5.914      Mean :23.532      Moderate Breeze:114007
## 3rd Qu.:6.210    3rd Qu.:27.478    Calm : 31116
## Max. :6.210     Max. :76.050    (Other) : 10497
## NA's :83453     NA's :84997     NA's : 83674
## depart_wind_direction_cat arrival_air_temperature arrival_dew_point
## E :107995      Min. : 1.00      Min. : -10.00
## ESE : 82685     1st Qu.:19.00    1st Qu.: 14.00
## SE : 74864      Median :23.00    Median : 18.00
## SSE : 66100     Mean :22.97     Mean : 17.27
## N : 61840       3rd Qu.:27.00    3rd Qu.: 21.00
## (Other):339155  Max. :41.00     Max. : 34.00
## NA's :112521    NA's :85460     NA's :86537
## arrival_relative_humidity arrival_wind_direction arrival_wind_speed
## Min. : 10.48    Min. : 0.0      Min. : 0.000
## 1st Qu.: 61.58  1st Qu.: 80.0   1st Qu.: 4.000
## Median : 74.44  Median :130.0   Median : 6.000
## Mean : 72.96   Mean :152.3     Mean : 6.796
## 3rd Qu.: 88.09  3rd Qu.:200.0   3rd Qu.: 9.000
## Max. :100.00    Max. :360.0     Max. :99.000
## NA's :86537     NA's :113150    NA's :85239
## arrival_sky_coverage arrival_pressure arrival_visibility
## Length:845160   Min. : 0.44     Min. :0.000
## Class :character 1st Qu.:29.91   1st Qu.:6.210
## Mode :character  Median :30.00   Median :6.210
##                  Mean :30.00   Mean :5.922
##                  3rd Qu.:30.09   3rd Qu.:6.210
##                  Max. :56.46    Max. :6.210
##                  NA's :85251     NA's :85010
## arrival_apparent_temperature arrival_wind_speed_scale
## Min. : -3.639    Light Breeze :243540
## 1st Qu.:19.000    Gentle Breeze :241538
## Median :23.000    Light Air :119946
## Mean :23.594     Moderate Breeze:112451
## 3rd Qu.:27.706    Calm : 31739
## Max. :65.511     (Other) : 10707
## NA's :86537      NA's : 85239
## arrival_wind_direction_cat
## E :103711
## ESE : 79476
## SE : 75207
## SSE : 66990
## S : 61992
## (Other):344634
## NA's :113150
```

A ideia é fazer a análise de cada variável individualmente.

Arrival

É o código do aeroporto de destino do voo.

```
table(bfd$arrival) %>%
  as.data.frame() %>%
  arrange(desc(Freq)) %>%
  rename(Aeroporto = Var1, Voos.Destino = Freq) %>%
```

```
head(10)
```

```
##      Aeroporto Voos.Destino
## 1      SBGR      115625
## 2      SBSP      73167
## 3      SBKP      63141
## 4      SBBR      51007
## 5      SBRJ      48757
## 6      SBCF      44666
## 7      SBRF      36595
## 8      SBPA      28063
## 9      SBSV      26545
## 10     SBCT      24619
```

Depart

É o código do aeroporto de origem do voo.

```
table(bfd$depart) %>%
  as.data.frame() %>%
  arrange(desc(Freq)) %>%
  rename(Aeroporto = Var1, Voos.Origem = Freq) %>%
  head(10)
```

```
##      Aeroporto Voos.Origem
## 1      SBGR      114683
## 2      SBSP      72997
## 3      SBKP      63171
## 4      SBBR      50735
## 5      SBRJ      48636
## 6      SBCF      44612
## 7      SBRF      36643
## 8      SBPA      28090
## 9      SBSV      26663
## 10     SBCT      24594
```

Route

É a rota voo com aeroporto de origem e aeroporto de chegada.

```
table(bfd$route) %>%
  as.data.frame() %>%
  arrange(desc(Freq)) %>%
  rename(Rota = Var1, Voos = Freq) %>%
  head(10)
```

```
##      Rota  Voos
## 1 SBSP-SBRJ 15203
## 2 SBRJ-SBSP 15189
## 3 SBPA-SBSP 6433
## 4 SBSP-SBPA 6432
## 5 SBCF-SBSP 6354
## 6 SBSP-SBCF 6351
## 7 SBSP-SBBR 6187
## 8 SBBR-SBSP 6183
## 9 SBPA-SBGR 5703
```

```
## 10 SBGR-SBPA 5575
```

Company

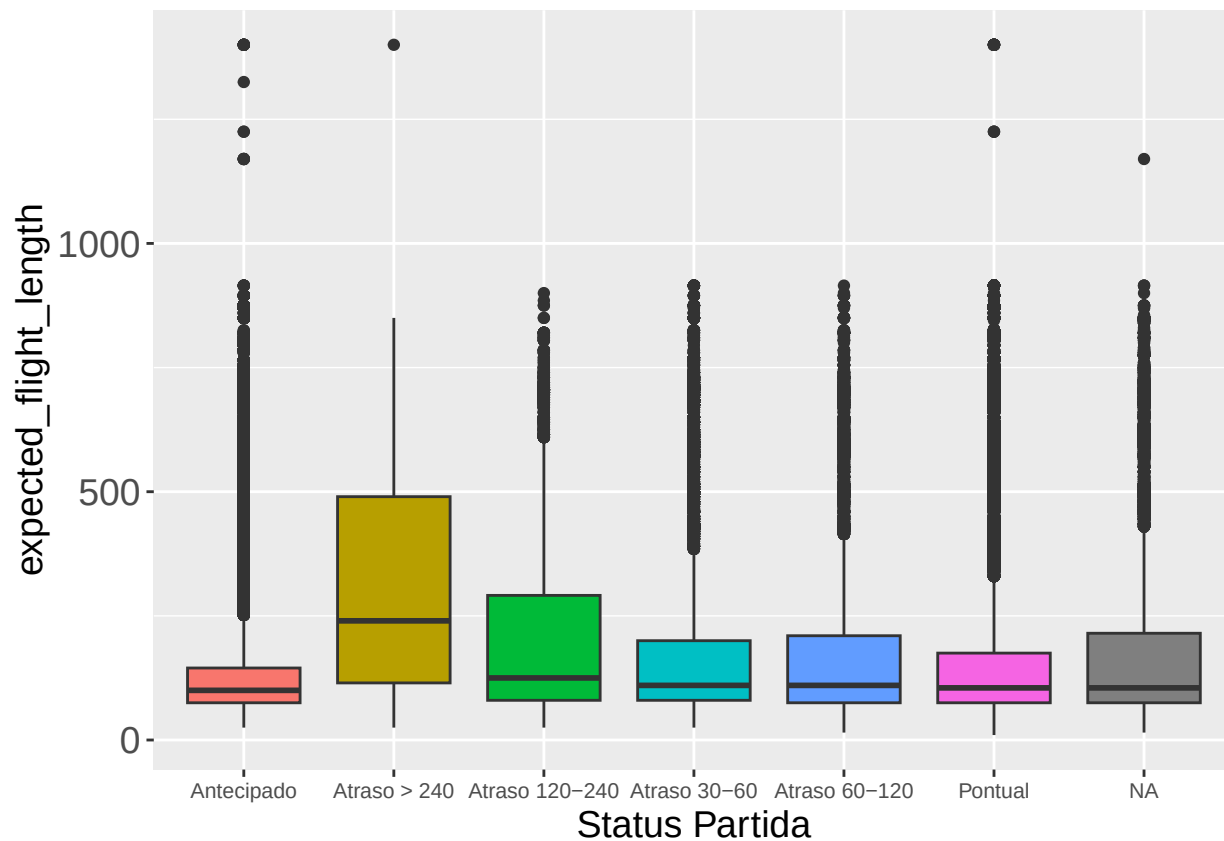
É a companhia aérea responsável pelo voo.

```
table(bfd$company) %>%  
  as.data.frame() %>%  
  arrange(desc(Freq)) %>%  
  rename(Companhia = Var1, Voos = Freq) %>%  
  head(10)
```

```
##      Companhia  Voos  
## 1          AZU 286730  
## 2          TAM 221577  
## 3          GLO 201210  
## 4          ACN 21285  
## 5          PTB 14117  
## 6          SID 8238  
## 7          TAP 7057  
## 8          CMP 6968  
## 9          ARG 6298  
## 10         LAN 4799
```

Expected Flight Length

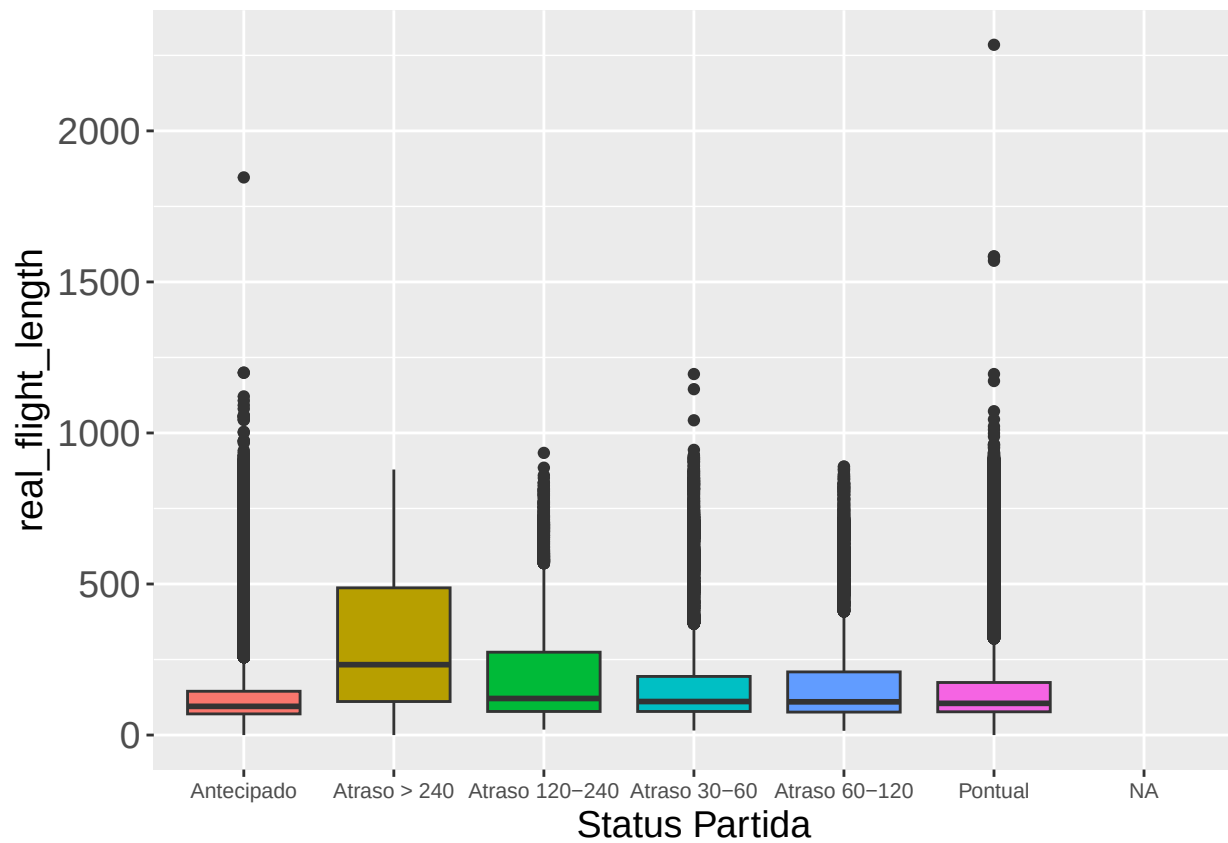
```
ggplot(bfd, aes(x=factor(status_depart),  
                y=expected_flight_length, fill=factor(status_depart))) +  
  xlab("Status Partida") +  
  theme(legend.position="none",  
        axis.text = element_text(size = 14),  
        text = element_text(size = 14),  
        axis.text.x = element_text(size = 8)) +  
  geom_boxplot()
```



Real Flight Length

```
ggplot(bfd, aes(x=factor(status_depart),
                  y=real_flight_length, fill=factor(status_depart))) +
  xlab("Status Partida") +
  theme(legend.position="none",
        axis.text = element_text(size = 14),
        text = element_text(size = 14),
        axis.text.x = element_text(size = 8)) +
  geom_boxplot()
```

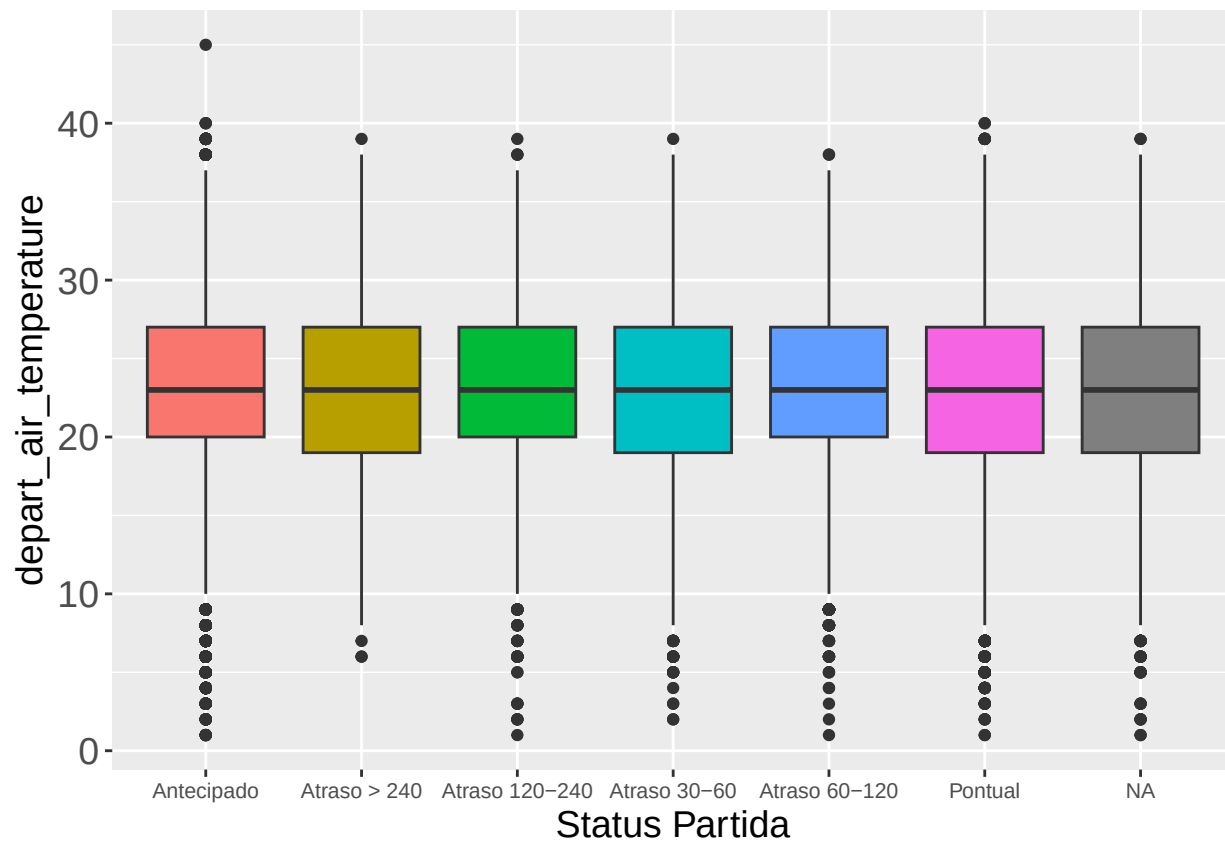
```
## Warning: Removed 34984 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



depart air temperature

```
ggplot(bfd, aes(x=factor(status_depart),
                  y=depart_air_temperature, fill=factor(status_depart))) +
  xlab("Status Partida") +
  theme(legend.position="none",
        axis.text = element_text(size = 14),
        text = element_text(size = 14),
        axis.text.x = element_text(size = 8)) +
  geom_boxplot()
```

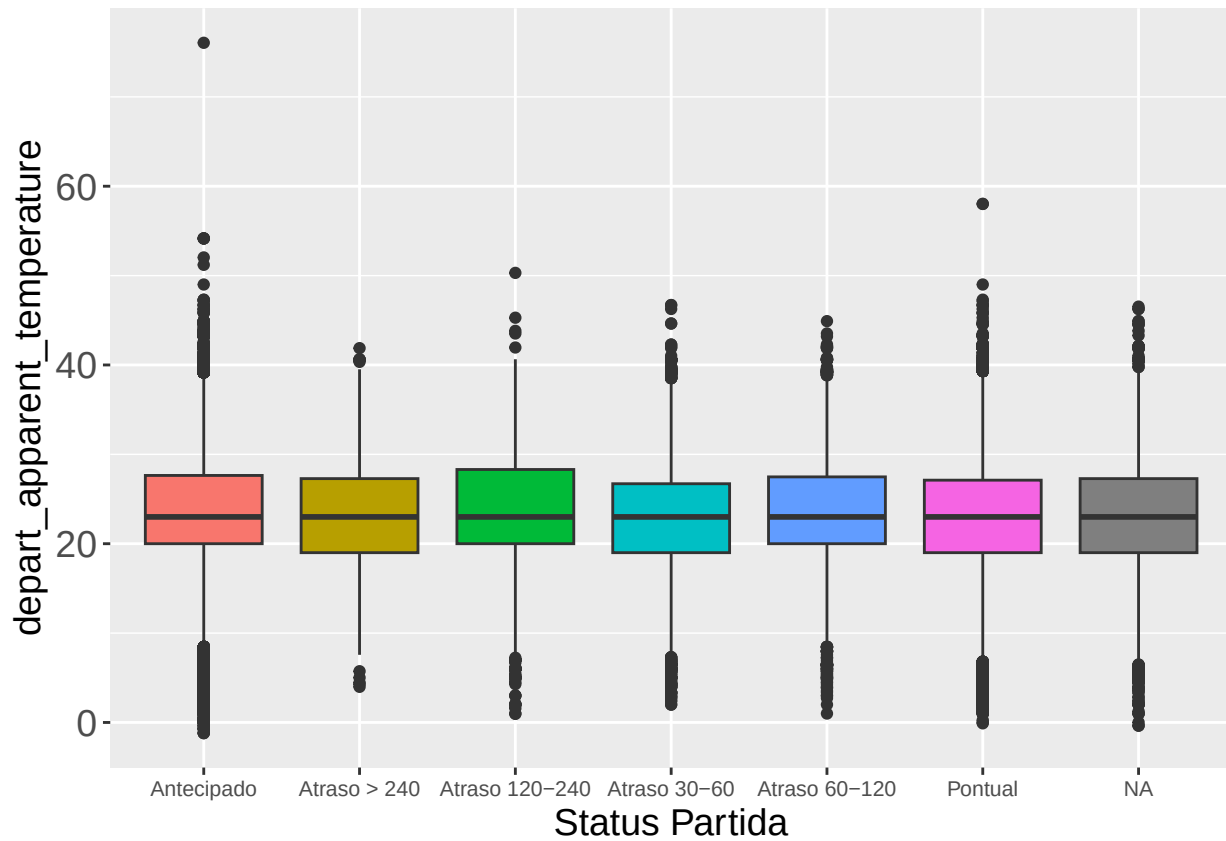
```
## Warning: Removed 83934 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



`depart_apparent_temperature`

```
ggplot(bfd, aes(x=factor(status_depart),
                  y=depart_apparent_temperature, fill=factor(status_depart))) +
  xlab("Status Partida") +
  theme(legend.position="none",
        axis.text = element_text(size = 14),
        text = element_text(size = 14),
        axis.text.x = element_text(size = 8)) +
  geom_boxplot()
```

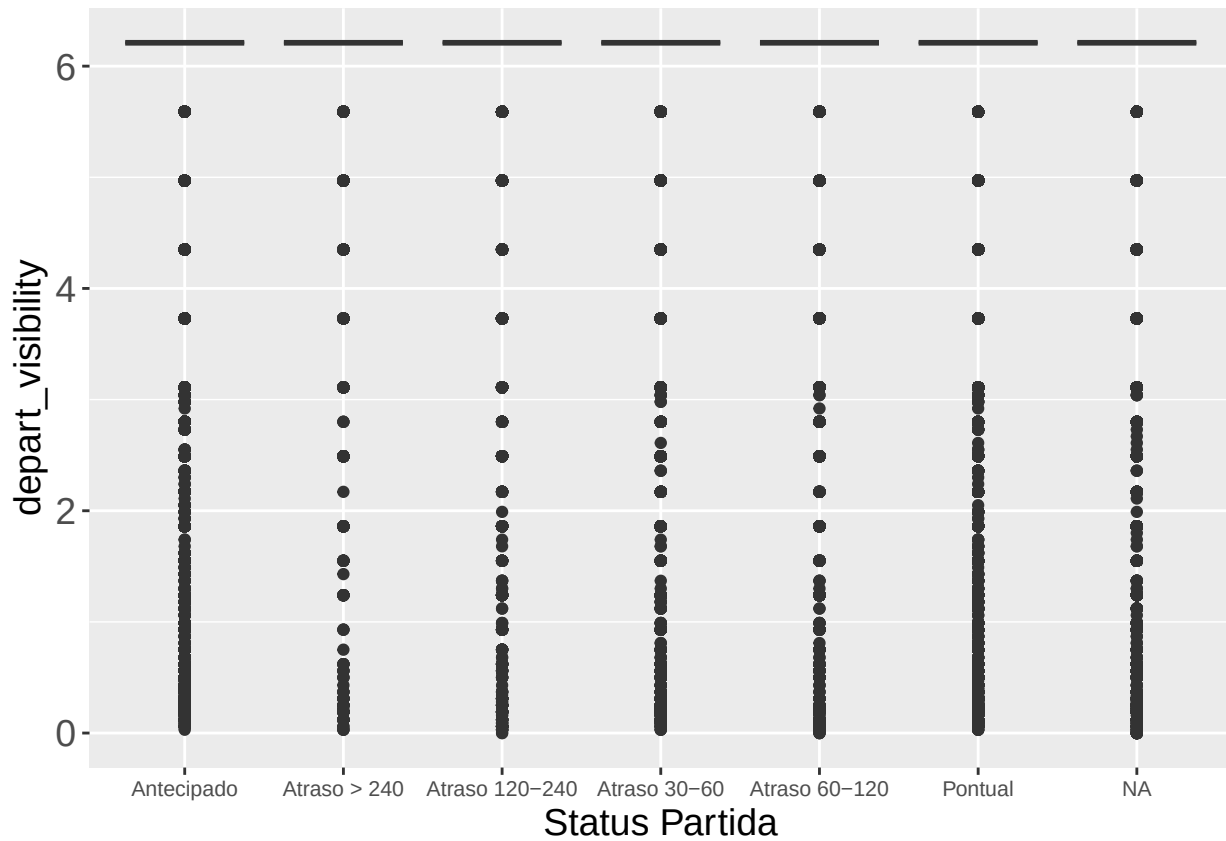
```
## Warning: Removed 84997 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

`depart_visibility`

```
ggplot(bfd, aes(x=factor(status_depart),
                  y=depart_visibility, fill=factor(status_depart))) +
  xlab("Status Partida") +
  theme(legend.position="none",
        axis.text = element_text(size = 14),
        text = element_text(size = 14),
        axis.text.x = element_text(size = 8)) +
  geom_boxplot()
```

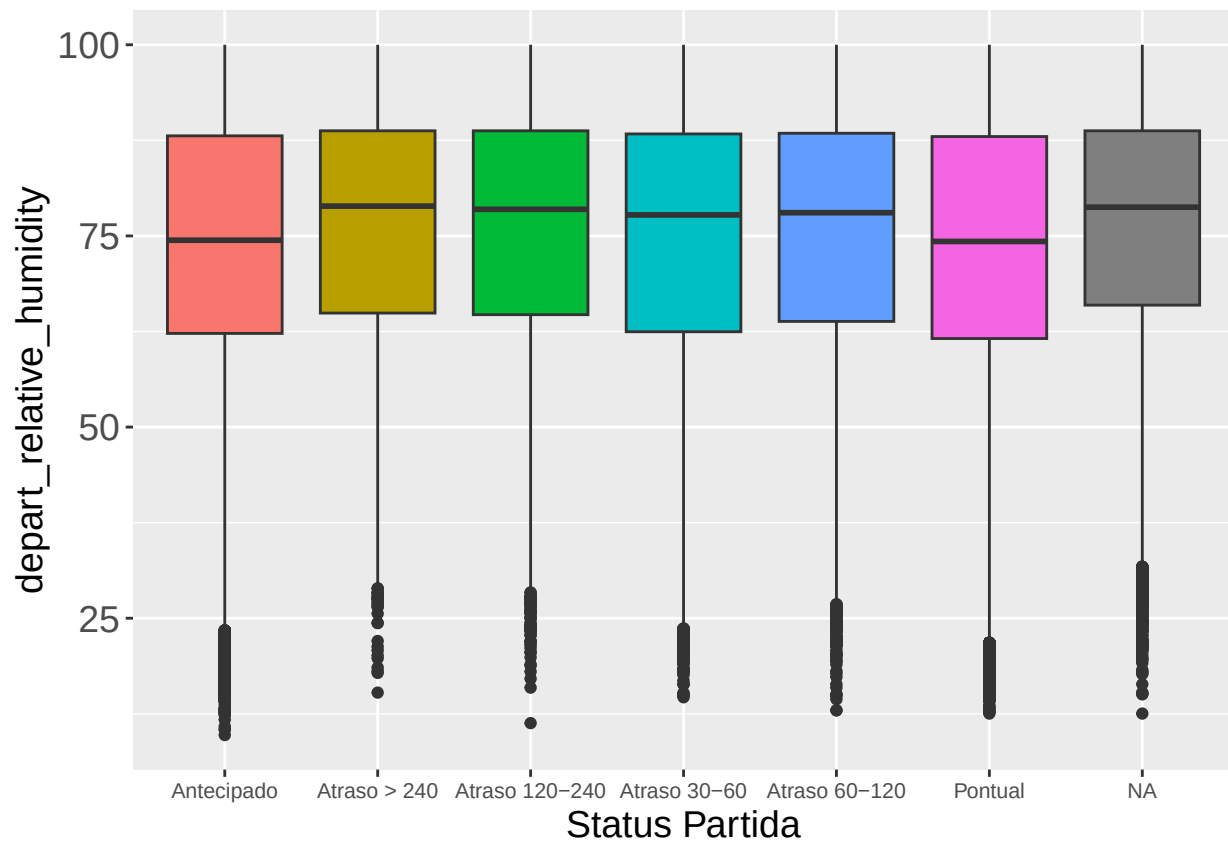
```
## Warning: Removed 83453 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



depart_relative_humidity

```
ggplot(bfd, aes(x=factor(status_depart),
                  y=depart_relative_humidity, fill=factor(status_depart))) +
  xlab("Status Partida") +
  theme(legend.position="none",
        axis.text = element_text(size = 14),
        text = element_text(size = 14),
        axis.text.x = element_text(size = 8)) +
  geom_boxplot()
```

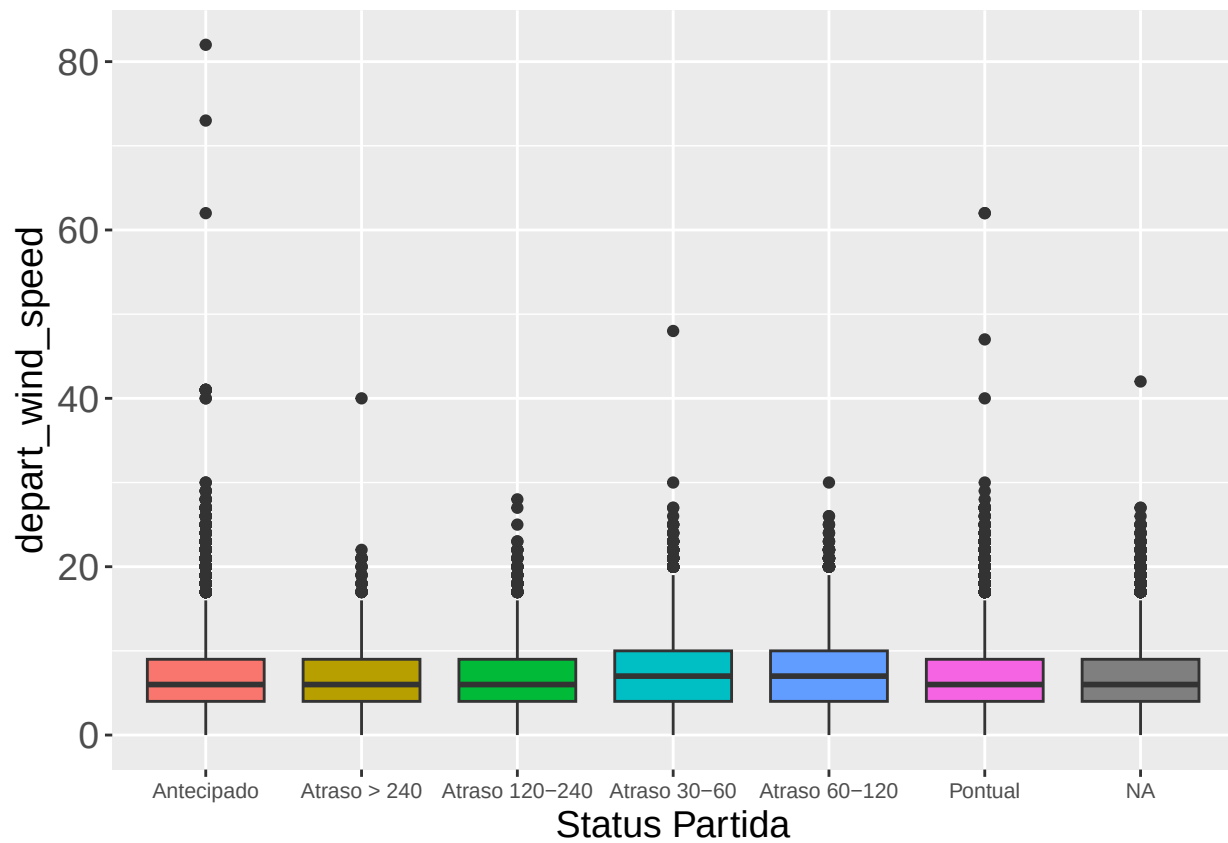
```
## Warning: Removed 84997 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



`depart_wind_speed`

```
ggplot(bfd, aes(x=factor(status_depart),
                  y=depart_wind_speed, fill=factor(status_depart))) +
  xlab("Status Partida") +
  theme(legend.position="none",
        axis.text = element_text(size = 14),
        text = element_text(size = 14),
        axis.text.x = element_text(size = 8)) +
  geom_boxplot()
```

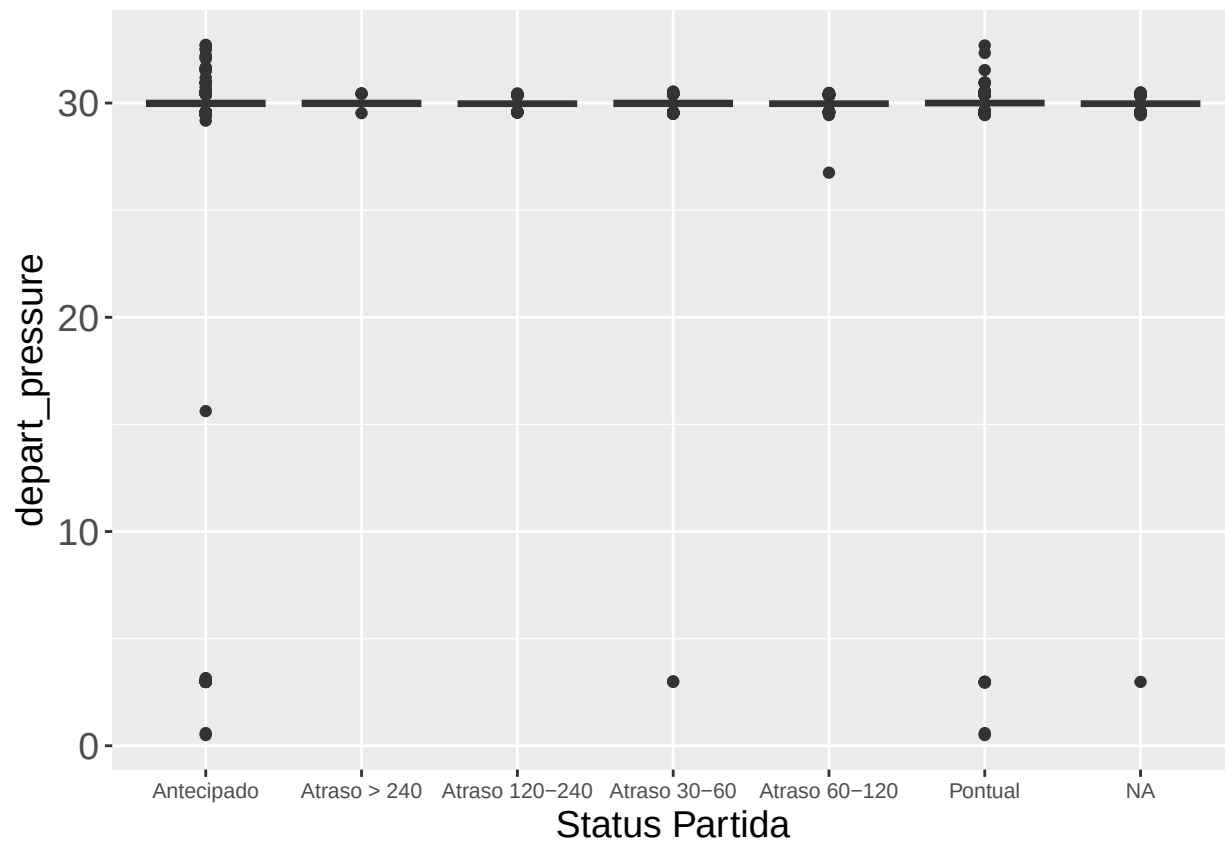
```
## Warning: Removed 83674 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



`depart__pressure`

```
ggplot(bfd, aes(x=factor(status_depart),
                    y=depart_pressure, fill=factor(status_depart))) +
  xlab("Status Partida") +
  theme(legend.position="none",
        axis.text = element_text(size = 14),
        text = element_text(size = 14),
        axis.text.x = element_text(size = 8)) +
  geom_boxplot()
```

```
## Warning: Removed 83718 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



depart_dew_point

```
ggplot(bfd, aes(x=factor(status_depart),
                  y=depart_dew_point, fill=factor(status_depart))) +
  xlab("Status Partida") +
  theme(legend.position="none",
        axis.text = element_text(size = 14),
        text = element_text(size = 14),
        axis.text.x = element_text(size = 8)) +
  geom_boxplot()
```

```
## Warning: Removed 84997 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

